LEEDS CITY COUNCIL

Highways and Transportation

Highways Services
St George House
40 Great George Street
Leeds LS1 3DL

Junction:	A64 York Road / Scholes Lane, Scholes, Leeds		
Company:	Telent	Controller Type:	Optima
Mains Supply:	230V / 50Hz	New / ~~Modification~~	
Operating Voltage:	ELV		
Engineer:	Dominic Dunne	Telephone No:	07566 759021

OTU SCN	26230	Cont SCN	26231
IP Address Information:			
Controller IP:	10.71.64.55	Gateway:	10.71.64.254
Mask:	255.255.255.0	OTU IP:	10.71.64.55

NOTE TO UTC PERSONNEL — This specification should be used for general purposes only.
The Signal Company documentation should be used for controller operation and timings unless amended on this sheet as RAM data.

RAM UPDATE (If no data then write NONE)	Reasons for change	Initials	Date
Write date current EPROM fitted:-			

ORN: 982L	Scheme Number: 716747	Issue: 1.0	Date: 20/10/2023

982L A64 York Road / Scholes Lane, Scholes, Leeds

Configuration Notes

General
A Telent 8-phase Optima ELV controller from contract 3664 shall provide the method of control in this specification.
The site is a 3-way Junction with puffin crossings on the south and east arms. Pedestrian detection is provided by Trafiones for kerbside detection and by AGDs for the on-crossing detection.
The normal method of control shall be MOVA.
Revert to QAR is to be initially enabled and have the ability to be switched off by timetable.
Trafiones to be addressed through 2 BPL2 card with IPs: 10.71.64.150 and 10.71.64.151

Quiescent All Red (QAR) – MOVA / VA control
In the absence of any demands and upon expiration of a damper timer (initially set to 10 seconds but alterable via the handset) the controller shall initiate a move to Quiescent All Red, stage 6, dummy DA, If under MOVA control, by way of a pulse to **MOVA det. 24.**

Kerbside Detector Operation.
The kerbside detectors will operate in 3 ways:
1. To eliminate the requirement for pedestrians to place a demand using the push button the kerbside detectors shall place an unlatched demand for pedestrian phase E & F, initially configured with a call delay of 2 seconds and a cancel period of 2 seconds (alterable by handset). This operation shall be configured with a software switch that can be switched by timetable and initially set to active.
2. The general operation of the kerbside and pushbutton demands will be as set out in TOPAS 2500. Pushbutton demands will be held and cancelled depending on the active state of the kerbside and any push button demand placed when no kerbside presence is detected shall place a latched demand.
3.The raw output from the kerbside detector shall set MOVA Det 42. This shall be used to extend the green man period through the use of priority links within MOVA.

NOTE: TBC - On DFM fail the kerbside detectors will not insert a demand for phase E & F and shall not set MOVA Det 42. At this time a pushbutton demand will be required.

If an unlatched demand is placed for phase E & F any active kerbside for that phase will hold the demand; the demand will only be dropped if all kerbsides associated with the phase are inactive.

Phase F & G Green Man Extensions
The pedestrian phase E & F green man period shall be capable of being extended. The raw detector state of kerbside detectors on phase E & F shall be used to apply VA extensions. The maximum green man period shall be governed by VA maxima values.

Audible Signal
An audible signal shall be present during the green man period with the facility to be switched by timetable. Initially set to turn off the audible signal everyday between 22:00 - 07:00.

Morwick Terrace - Demand Unit
An additional push button unit shall be installed adjacent to the driveway of Morwick Terrace. In UTC/CLF/VA this shall place a demand for Phase H (Dummy DB, stage 5) - to provide an all red period for a safe egress due to the inability to safely signal the approach. In MOVA it shall pulse MOVA Det 24.

Phase Terminations
Phase A & B to always open and close together. A & D to terminate together at the end of stage 2.

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Configuration Notes

Mova Conditioning

MOVA UTC Interface:

Control: F1 F2 F3 F4 F5 MTO DS1 DS2 DS3 DS4
Reply: G1 G2 G3 G4 G5 CRB MO
Wait Lamp Mimic: Whilst pedestrian phase wait lamp is lit, input to MOVA:
Ped E = MOVA Det 22
Ped F = MOVA Det 23
Ped H (Morwick Terrace AR demand) = MOVA Det 21

CRB Toggle: CRB configured to toggle every 120 seconds if MOVA is not in control.

UTC & MOVA confirmation LED

Within the manual panel the UTC control **confirmation LED shall FLASH when the controller is under MOVA control and remain on solid when under UTC control**.
A clearly labelled, manually selected VA switch shall be provided within the manual panel which shall override MOVA control when active.

MOVA Loop Positions

Detectors shall be wired into packs as shown below, to prevent chattering.

Approach	Loop	Dist From Stop Line (m)	Longitudinal Dimension (m)	Shape	Clearance Distance		Detector Pack No.
					N/S	O/S	
A64 York Road Eastbound	AIN1	90	1.7	Diamond	0.4	0.8	1
	AX2	45	1.7	Diamond	0.4	0.8	1
	ASL3	1	2	Rectangular	0.4	0.8	1
	AACSL17	Virtual					
	DIN4	84	1.7	Diamond	0.8	0.8	1
	DX5	39	1.7	Diamond	0.4	0.8	2
	DSL6	1	2	Rectangular	0.8	0.4	2
	EACSL18	Virtual					
A64 York Road Westbound	BIN7	90	1.7	Diamond	0.4	0.8	2
	BX8	45	1.7	Diamond	0.4	0.8	2
	BX9	45	1.7	Diamond	0.8	0.4	3
	BSL10	1	2	Rectangular	0.4	0.8	3
	BSL11	1	2	Rectangular	0.8	0.4	3
	BACSL19	Virtual					
Scholes Lane	CIN12	80	1.7	Diamond	0.4	0.8	3
	CX13	40	1.7	Diamond	0.4	0.8	4
	CX14	35	1.7	Diamond	0.8	0.8	4
	CSL15	1	2	Rectangular	0.4	0.8	4
	CSL16	1	2	Rectangular	0.8	0.8	4
	CACSL20	Virtual					

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Configuration Notes

<u>Flow & Occupancy Loops</u>
The following conditioning shall be subject to individual software switches, allowing the logic to be switched on or off.
After every 4 vehicles that pass over AIN1, change the state of reply bit 29 (FL1) (SCN 43618) ie. in reply or not.
Every time a vehicle is detected over AIN1, pulse reply bit 25 (OC1). (SCN 43618)
After every 4 vehicles that pass over DIN4, change the state of reply bit 32 (FL4) (SCN 43621) ie. in reply or not.
Every time a vehicle is detected over DIN4, pulse reply bit 28 (OC4). (SCN 43621)
After every 4 vehicles that pass over BIN7, change the state of reply bit 30 (FL2) (SCN 43619) ie. in reply or not.
Every time a vehicle is detected over BIN7, pulse reply bit 26 (OC2). (SCN 43619)
After every 4 vehicles that pass over CIN12, change the state of reply bit 31 (FL3) (SCN 43620) ie. in reply or not.
Every time a vehicle is detected over CIN12, pulse reply bit 27 (OC3). (SCN 43620)

<u>Corridor Control Strategy</u>
The following control bits are processed by UTC and sent based on different traffic conditions on the corridor. The purpose of these is to hold vehicle green to flush traffic through when required.

If control bit 21 (SF1) is received, pulse MOVA Det 30 for 2 seconds. (SCN 43610)
If control bit 22 (SF2) is received, pulse MOVA Det 31 for 2 seconds. (SCN 43611)
If control bit 23 (SF3) is received, pulse MOVA Det 32 for 2 seconds. (SCN 43612)
If control bit 24 (SF4) is received, pulse MOVA Det 33 for 2 seconds. (SCN 43613)

<u>Dynamic Dataset Changes</u> - Based on the flow/occupancy/queueing conditions of the corridor it is desirable to change MOVA Datasets. The following logic and MOVA detectors facilitate the changes.

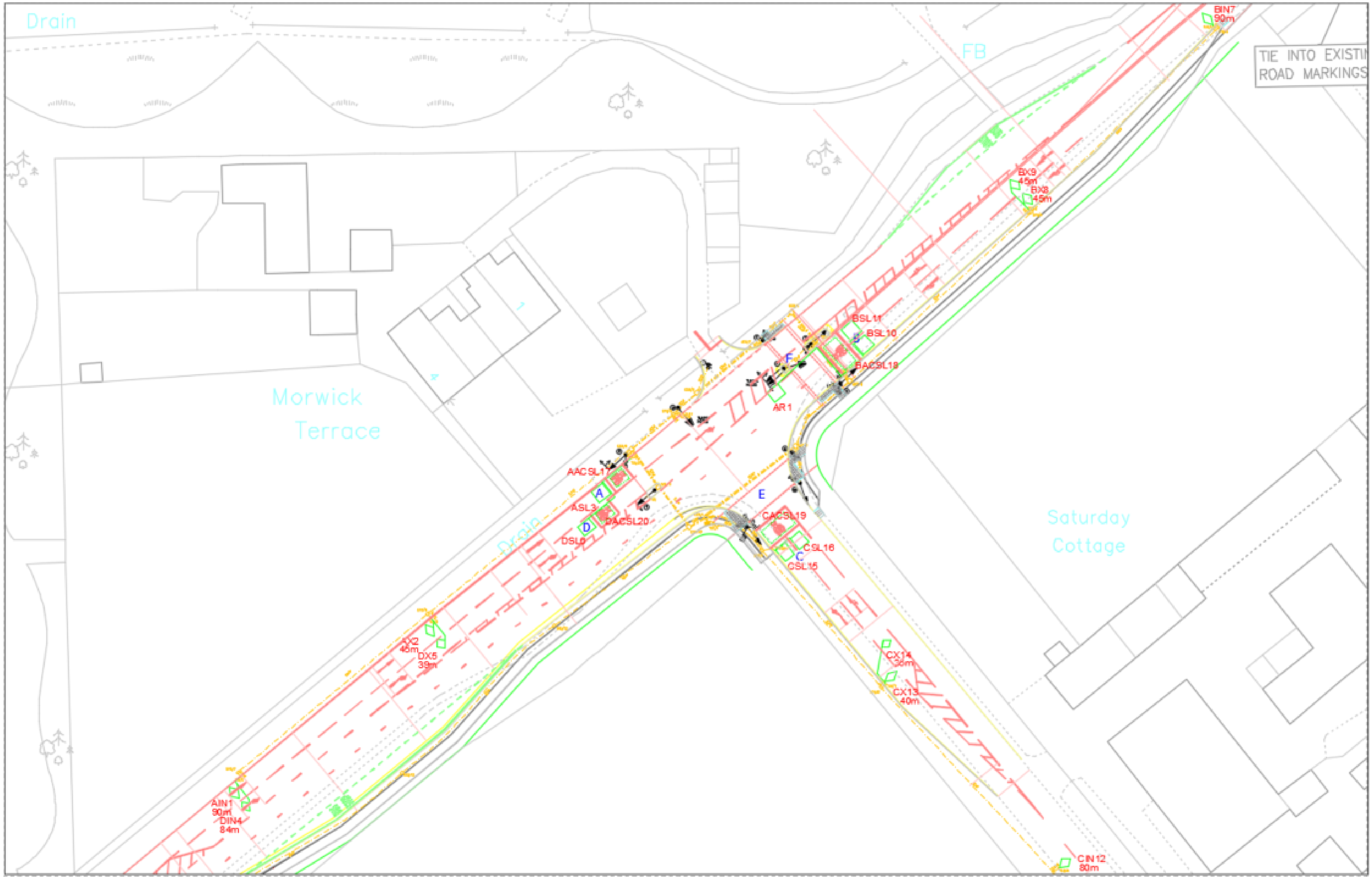
If control bit 17 (DS1) is being received from UTC, output to MOVA Det.26 (SCN 43614)
If control bit 18 (DS2) is being received from UTC, output to MOVA Det.27 (SCN 43615)
If control bit 19 (DS3) is being received from UTC, output to MOVA Det.28 (SCN 43616)
If control bit 20 (DS4) is being received from UTC, output to MOVA Det.29 (SCN 43617)

<u>Bus Priority</u>
Implemented via RTIG Bus Triggers shall be configured and input to MOVA dets 46 through to 54 respectively.

<u>HGV Priority</u>
RTEM profile detection** to monitor vehicles passing over AIN1, BIN3 and BIN4, and pulse MOVA det.39 and 40 respectively for 2 seconds.

(Note!** RTEM not implemented at present but leave MOVA det.39 and 40 assigned to HGV)

Site Layout



NOT TO SCALE

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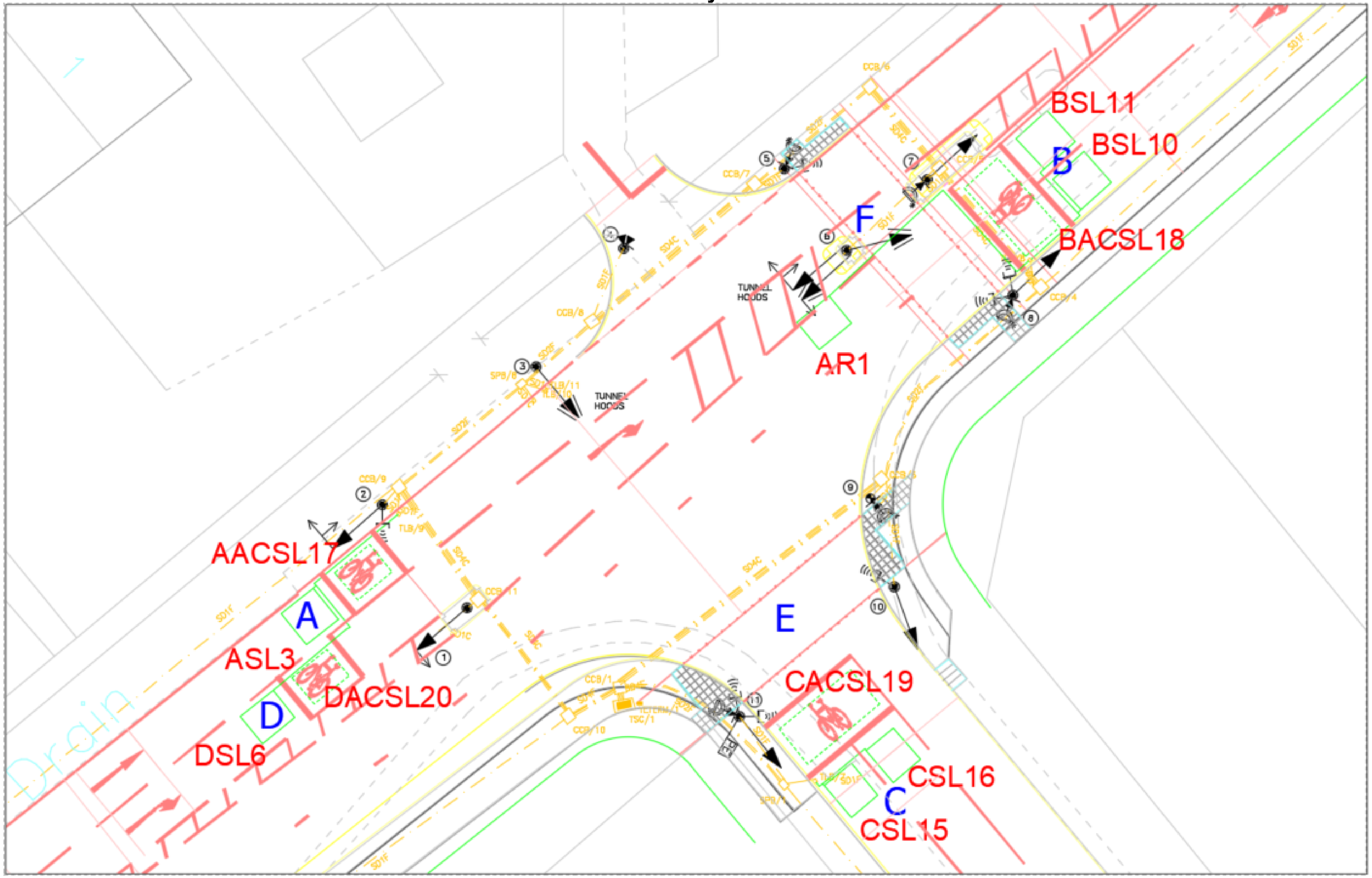
Site Layout



NOT TO SCALE

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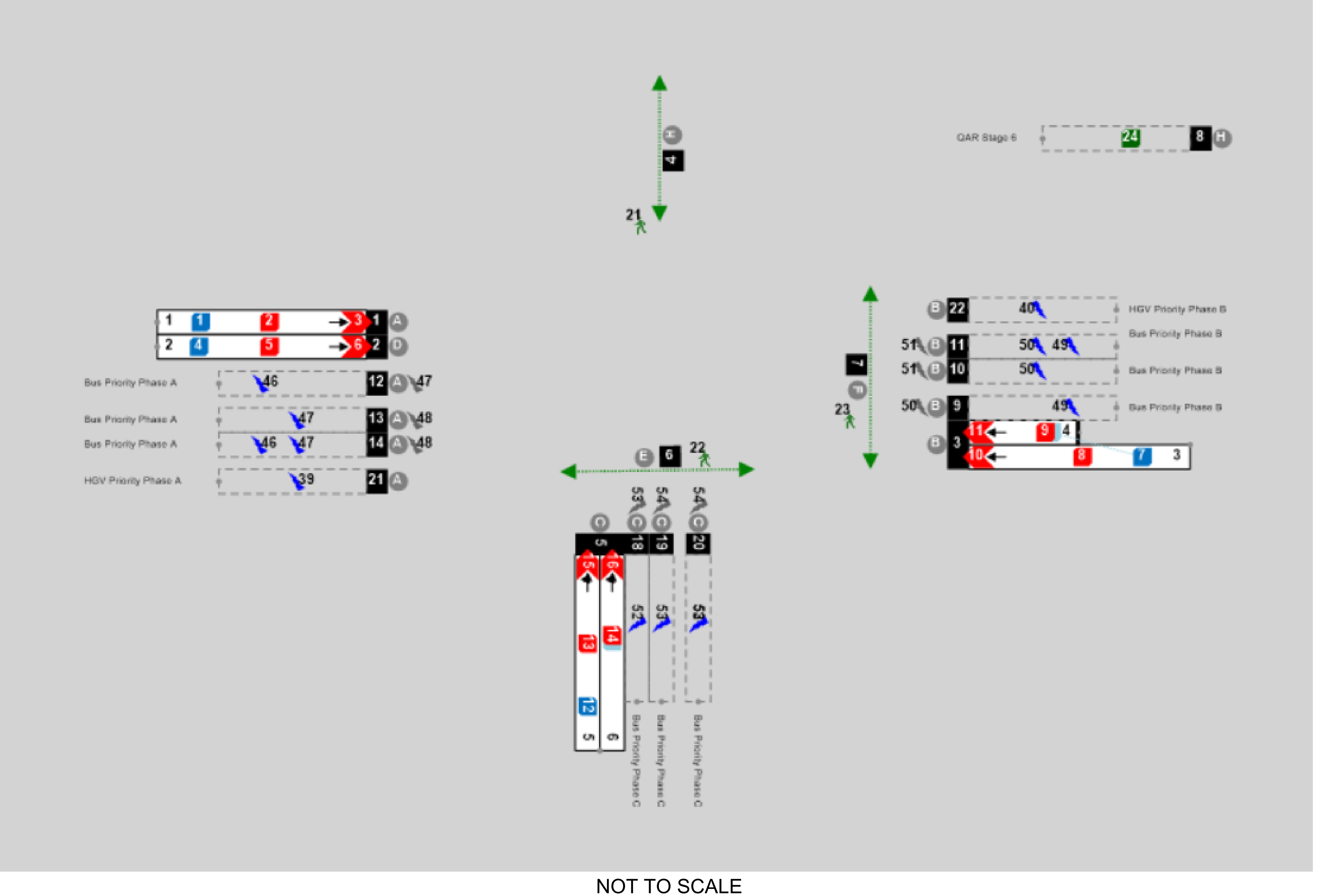
MOVA Link Diagram



NOT TO SCALE

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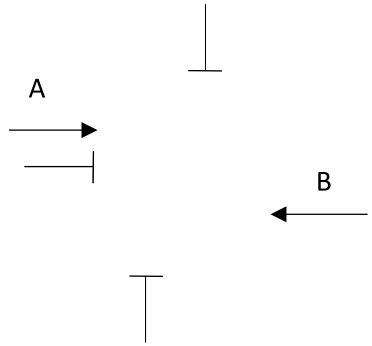
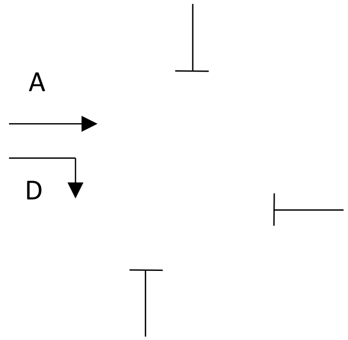
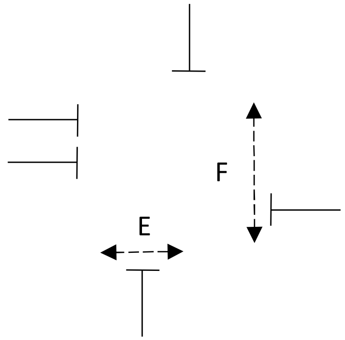
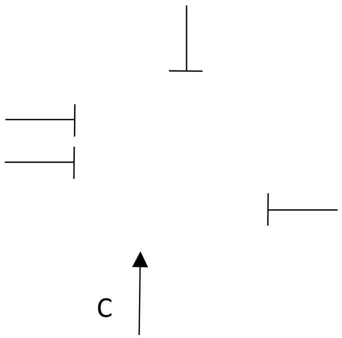
BASIC STAGE DATA FOR STREAM NO 1			
Stage: 0	Stage: 1	Stage: 2	Stage: 3
MANUAL ALL RED			
Stage: 4	Stage: 5	Stage:	Stage:
	QAR DUMMY G (DA) DUMMY H (DB)		

	STREAM 1	STREAM 2	STREAM 3	STREAM 4
ALL RED STAGE	0			
START UP STAGE	1			
REVERSION STAGE	5			

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USE OF PHASES		PHASE	APPEAR	TERMINATION					MAXIMUMS			
									(Day)	(AM)	(PM)	(Night)
		TYPE	TYPE	TYPE	ASSOC	PED	PERIOD	MIN	SET	SET	SET	SET
	LOCATION	NOTE 3	NOTE 1	NOTE 2	PHASE	BLKT	V	GREEN	A	B	C	D
A	A64 York Road Eastbound	T	0	0				7	30	35	50	30
B	A64 York Road Westbound	T	0	0				7	30	50	35	30
C	Scholes Lane	T	0	0				7	15	20	20	15
D	A64 York Road Eastbound RT	T	0	0				7	12	15	15	12
E	Puffin across Scholes Lane	PU	0	0			3	5	8	8	8	8
F	Puffin across York Road	PU	0	0		-	3	5	8	8	8	8
G	Dummy (DA) Quiescent All Red	D	0	0				2				
H	Dummy (DB) Morwick Terrace AR	D	0	0				7				

Note 1:- Condition of Appearance

0 - Phase always appears when stage runs

1 - Phase only appears if demand exists at start of inter-stage

2 - Phase appears, if demanded, at any time up until the end of the stage in which the phase runs

3 - Phase appears, if demanded at any time during the stage up until the window time expires

Note 2:- Condition of Termination

0 - Terminates at end of stage

1 - Terminates when associated phase gains right of way

2 - Terminates when associated phase loses right of way

3 - Terminates at end of minimum green

4 - Terminates at end of maximum green

5 - Terminates subject to special conditioning

Note 3: - Phase Types

T - Traffic

PD- Pedestrian

PU- Puffin

TN - Toucan Nearside

TF - Toucan Farside

F - Filter Arrow

I - Indicative Arrow

D - Dummy

L - L.R.T

S - Switched Sign

C - Cycle

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PUFFIN CROSSING TIMINGS								
Phase	Crossing Width (m)	Comfort Factor	Period III	Period IV	Period V	Period VI	Period VII	Period VIII
E	13.9	0	3	5	3	14	0	0
F	13.4	0	3	5	3	14	0	0

USE OF STAGES - STAGES/PHASE RELATIONSHIP																							
STAGE	A	B	C	D	E	F	G	H															
0							M	A	N	U	A	L		A	L	L		R	E	D			
1	✓	✓																					
2	✓			✓																			
3					✓	✓																	
4			✓																				
5							✓	✓		Q	A	R											

STREAM	WINDOW TIME
1	
1	
1	
1	
1	
1	

MAX REVERTIVE DEMAND PHASE																							

PHASES	A	B	C	D	E	F	G	H
A			C			C	C	C
B			C	C	C	C	C	C
C	C	C		C	C	C	C	C
D		C	C		C		C	C
E		C	C	C			C	C
F	C	C	C				C	C
G	C	C	C	C	C	C		
H	C	C	C	C	C	C		

CONFLICTING / OPPOSING PHASES

C = CONFLICTING & OPPOSING

O = OPPOSING ONLY

. = NO CONFLICT

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PHASES		PHASE INTERGREEN TIMES						
	A	B	C	D	E	F	G	H
A		.	6	.	.	10	3	9
B	.		8	6	9	6	3	8
C	6	6		6	6	11	3	8
D	.	7	8		11	.	3	8
E	.	5*	5*	5*		.	3	5*
F	5*	5*	5*	.	.		3	5*
G	2	2	2	2	2	2		.
H	5	5	8	6	7	7	.	

STARTING INTERGREEN PERIOD	12

* = see Puffin Timings sheet

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PHASES	A	B	C	D	E	F	G	H
A		.	5	.	.	8	3	7
B	.		6	5	7	5	3	6
C	5	5		5	5	9	3	6
D	.	5	6		9	.	3	6
E	.	5	5	5		.	3	5
F	5	5	5	.	.		3	5
G	2	2	2	2	2	2		.
H	5	5	6	5	5	5	.	

PHASE INTERGREEN HANDSET LIMIT VALUES

MIN GREEN VALUE	
LOW	HIGH
7	
7	
7	
7	
5	
5	
2	
7	

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ADDITIONAL PHASE DELAYS														
No	Delay Phase	Change From	To Stage	By (X) Secs	No	Delay Phase	Change From	To Stage	By (X) Secs	No	Delay Phase	Change From	To Stage	By (X) Secs
1	A	5	2	1*	29					57				
2					30					58				
3					31					59				
4					32					60				
5					33					61				
6					34					62				
7					35					63				
8					36					64				
9					37					65				
10					38					66				
11					39					67				
12					40					68				
13					41					69				
14					42					70				
15					43					71				
16					44					72				
17					45					73				
18					46					74				
19					47					75				
20					48					76				
21					49					77				
22					50					78				
23					51					79				
24					52					80				
25					53					81				
26					54					82				
27					55					83				
28					56					84				

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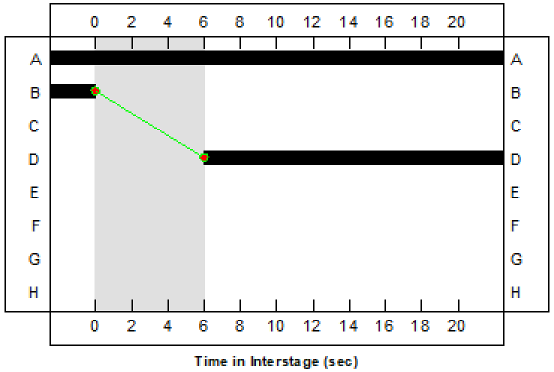
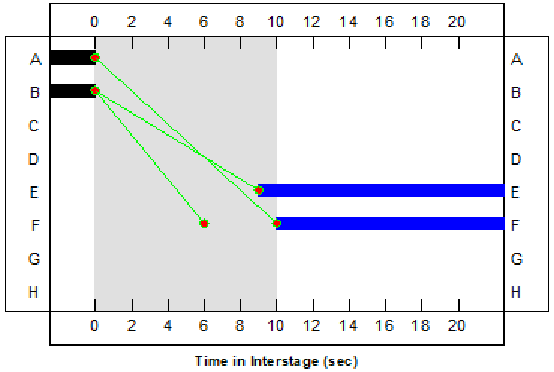
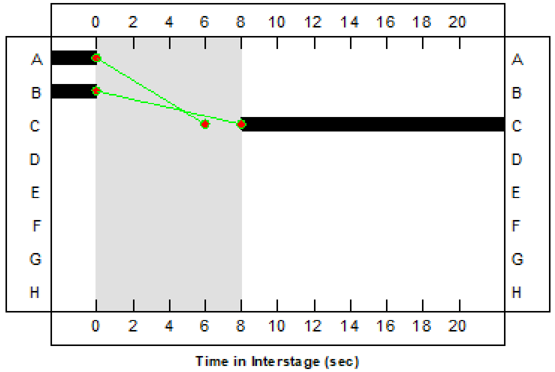
Stage Change 1 -> 2



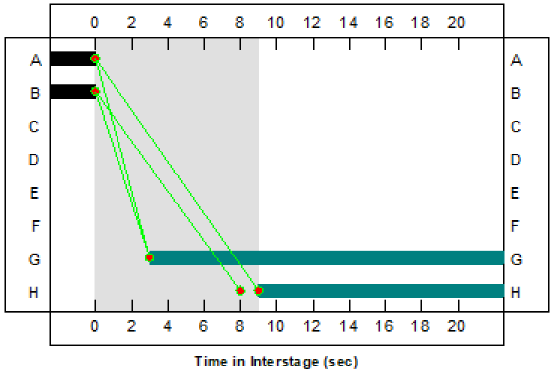
Stage Change 1 -> 3



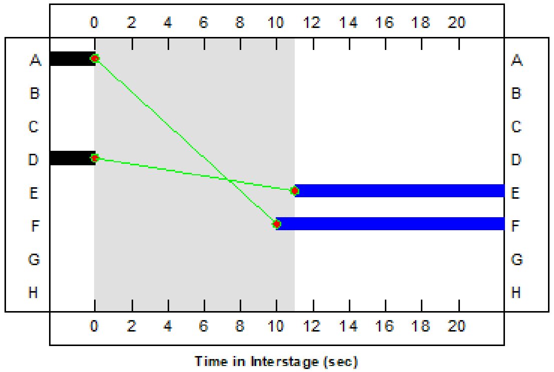
Stage Change 1 -> 4



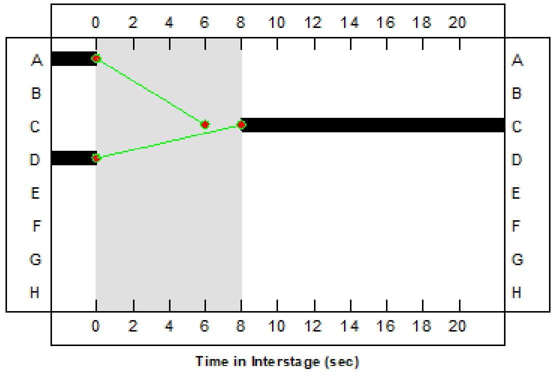
Stage Change 1 -> 5



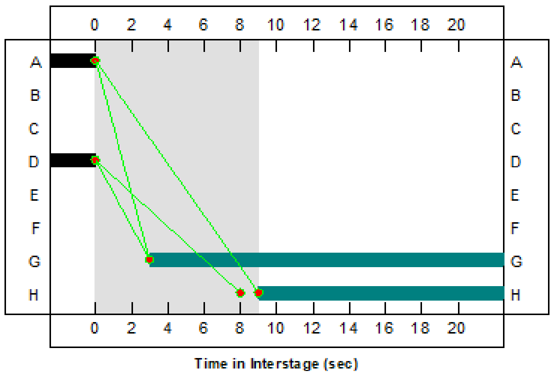
Stage Change 2 -> 3



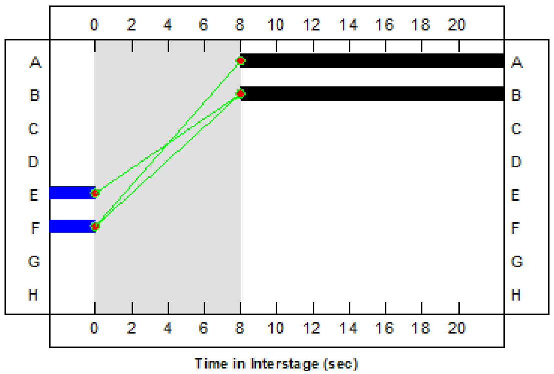
Stage Change 2 -> 4



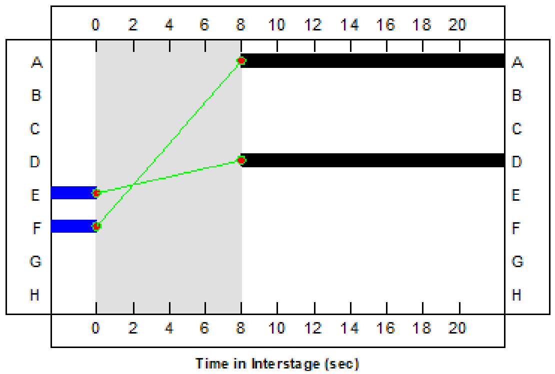
Stage Change 2 -> 5



Stage Change 3 -> 1



Stage Change 3 -> 2



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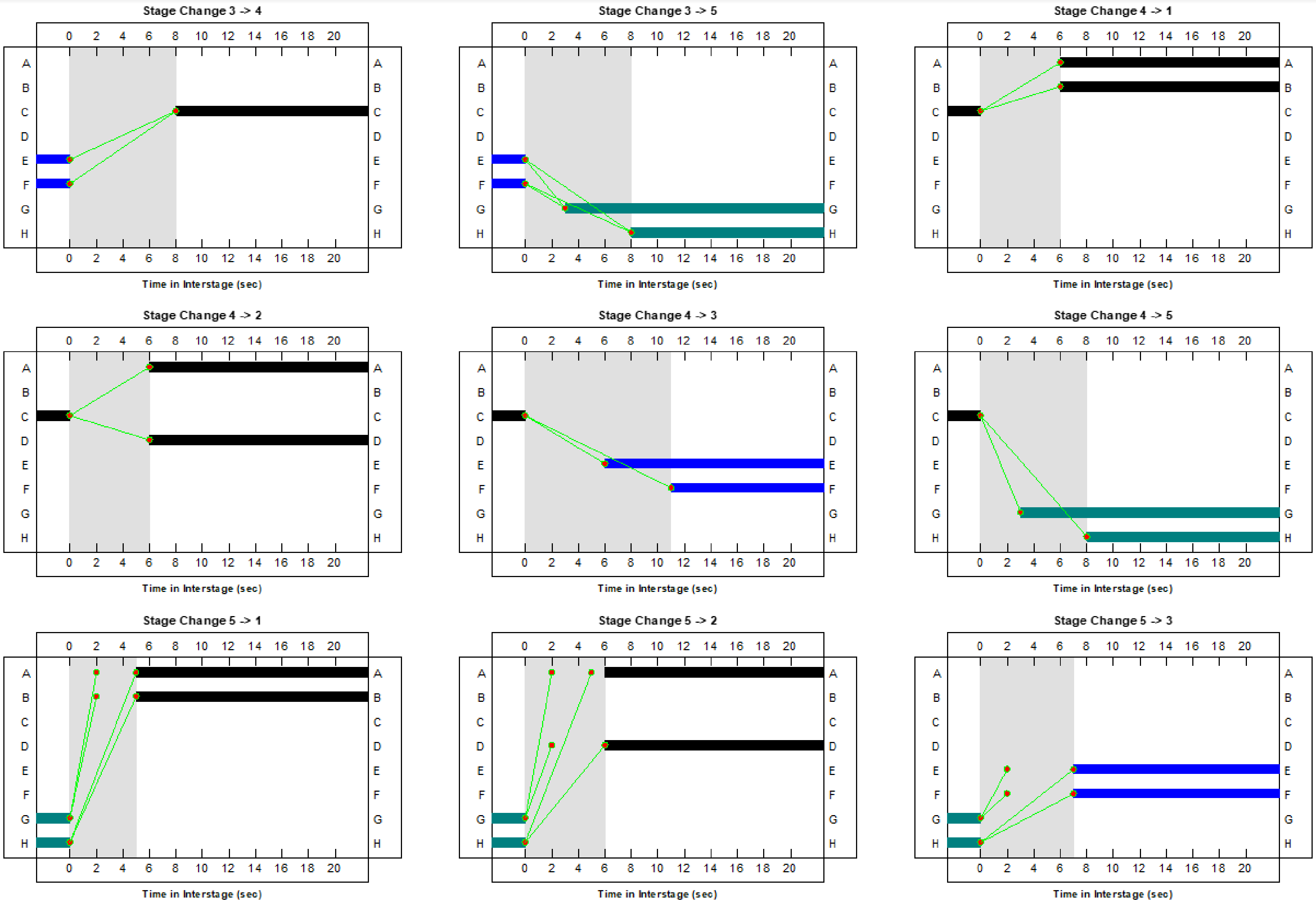

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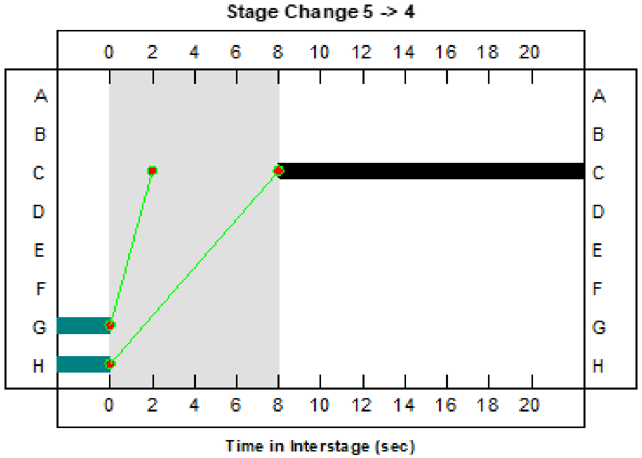

(Insert intergreen diagrams here)

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PROHIBITED / ALTERNATIVE STAGE MOVEMENTS UP TO 16 STAGES

FROM	TO STAGE															
STAGE	0	1	2	3	4	5										
0		✓	✓	✓	✓	✓										
1	✓		✓	✓	✓	✓										
2	✓	5		✓	✓	✓										
3	✓	✓	✓		✓	✓										
4	✓	✓	✓	✓		✓										
5	✓	✓	✓	✓	✓											

TICK AS REQUIRED		
MODE	These Restrictions are for	No Restrictions
Urban Traffic Control	✓	
Cableless Linking	✓	
MOVA	✓	
Hurry Call	✓	
Manual	✓	
Priority	✓	
Emergency	✓	

NOTE:

✓	=	Permitted
P	=	Prohibited
X	=	Ignore
N'	=	ALTERNATIVE MOVES where 'N' is number of stage to move to

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DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS													
ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
1	AIN1	V		A				4.6			1	Y	MOVA Det 1
2	AX2	V	A	A				4.6			1	A	MOVA Det 2
3	ASL3	V	A	A		3		0.6			1	A	MOVA Det 3
4	DIN4	V		E				4.6			1	Y	MOVA Det 4
5	DX5	V	E	E				4.6			1	A	MOVA Det 5
6	DSL6	V	B	B		3		0.6			1	A	MOVA Det 6
7	BIN7	V		B				4.6			1	Y	MOVA Det 7
8	BX8	V	B	B				4.6			1	A	MOVA Det 8
9	BX9	V	B	B				4.6			1	A	MOVA Det 9
10	BSL10	V	B	B		3		0.6			1	A	MOVA Det 10
11	BSL11	V	B	B		3		0.6			1	A	MOVA Det 11
12	CIN12	V		C				4.2			1	Y	MOVA Det 12
13	CX13	V	C	C				4.2			1	A	MOVA Det 13
14	CX14	V	C	C				3.6			1	A	MOVA Det 14
15	CSL15	V	C	C		3		0.6			1	A	MOVA Det 15
16	CSL16	V	C	C		3		0.6			1	A	MOVA Det 16
17	AACSL18	V	A	A				0.6			1	Y	MOVA Det 17
18	BACSL19	V	B	B				0.6			1	Y	MOVA Det 18
19	CACSL20	V	C	C				0.6			1	Y	MOVA Det 19
20	EACSL21	V	E	E				0.6			1	Y	MOVA Det 20
21	PBU H1 - Pole 4	V	H								3	Y	MOVA Det 21

✓ Tick appropriate boxes

Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	10	24
3	10	0
4	10	120

Detector Types:-

V	Vehicle	**C**	Clearance	**SU**	Speed Upstream
PB	Pushbutton	**Q**	Queue	**SD**	Speed Downstream
K	Kerbside	**U1**	Uni Directional 1		
O	On-Crossing	**U2**	Uni Directional 2		

Error States

- **A** Fail Active
- **I** Fail Inactive
- **Y** Fail in input state

Paired Detector:-

Specify ID of paired Uni-directional or kerbside/pushbuttons

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DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS													
ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
22	AR1	V						2			1	A	See All Red sheet
23	PBU G - Pole 5	PB	G								3	Y	
24	Kerbside - Pole 5	K	G						23	1	2	I	
25	On Crossing - Pole 5	O						0.5			1	A	
26	PBU G - Pole 7	PB	G								3	Y	
27	PBU G - Pole 8	PB	G								3	Y	
28	Kerbside - Pole 8	K	G						27	1	2	I	
29	On Crossing - Pole 8	O						0.5			1	A	
30	PBU F - Pole 9	PB	F								3	Y	
31	Kerbside - Pole 10	K	F						30	1	2	I	
32	On Crossing - Pole 10	O						0.5			1	A	
33	PBU F - Pole 11	PB	F								3	Y	
34	Kerbside - Pole 11	K	F						33	1	2	I	
35	On Crossing - Pole 11	O						0.5			1	A	
36	Wait Confirm F												MOVA Det 22
37	Wait Confirm G												MOVA Det 23
38	Tactile Fault - Pole 5										4	A	
39	Tactile Fault - Pole 7										4	A	
40	Tactile Fault - Pole 8										4	A	
41	Tactile Fault - Pole 9										4	A	
42	Tactile Fault - Pole 11										4	A	

✓ Tick appropriate boxes

Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	10	24
3	10	0
4	10	120

Detector Types:-

V	Vehicle	**C**	Clearance	**SU**	Speed Upstream
PB	Pushbutton	**Q**	Queue	**SD**	Speed Downstream
K	Kerbside	**U1**	Uni Directional 1		
O	On-Crossing	**U2**	Uni Directional 2		

Error States

A	Fail Active
I	Fail Inactive
Y	Fail in input state

Paired Detector:-

Specify ID of paired Uni-directional or kerbside/pushbuttons

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MOVA DETECTORS								
No.	Name / Description	Action	No.	Name / Description	Action	No.	Name / Description	Action
1	AIN1		26	DS1	MOVA Dataset 1	51	Bus Trigger Cancel B	Cancels Bus Priority on B
2	AX2		27	DS2	MOVA Dataset 2	52	Bus Trigger Call C	Calls Bus Priority on C
3	ASL3		28	DS3	MOVA Dataset 3	53	Bus Trigger Hold C	Holds Bus Priority on C
4	DIN4		29	DS4	MOVA Dataset 4	54	Bus Trigger Cancel C	Cancels Bus Priority on C
5	DX5		30	SF1	Control Strategy 1	55		
6	DSL6		31	SF2	Control Strategy 2	56		
7	BIN7		32	SF3	Control Strategy 3	57		
8	BX8		33	SF4	Control Strategy 4	58		
9	BX9		34			59		
10	BSL10		35			60		
11	BSL11		36			61		
12	CIN12		37			62		
13	CX13		38			63		
14	CX14		39	AIN1	HGV Priority A	64		
15	CSL15		40	BIN4	HGV Priority B	65		
16	CSL16		41			66		
17	AACSL17		42	Kerbside	Green man extension	67		
18	BACSL18		43	On Crossing	QAR extension	68		
19	CACSL19		44			69		
20	DACSL20		45			70		
21	PBU H	Demands H (Stage 5)	46	Bus Trigger Call A	Calls Bus Priority on A	71		
22	PBU E	Demands E	47	Bus Trigger Hold A	Holds Bus Priority on A	72		
23	PBU F	Demands F	48	Bus Trigger Cancel A	Cancels Bus Priority on A	73		
24	All Red Revert	QAR (Stage 5)	49	Bus Trigger Call B	Calls Bus Priority on B	74		
25	AR1		50	Bus Trigger Hold B	Holds Bus Priority on B	75		

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ADDITIONAL SWITCHES & LED'S (ON MANUAL PANEL)		
POSITION	USE	MOMENTARY *
SWITCH 1		
LED 1	MOVA Enabled (Flashing) UTC Mode (Solid)	
SWITCH 2		
LED 2		
SWITCH 3		
LED 3		
HURRY CALL AUX 4		
HIGHER PRIORITY AUX 5		

HURRY CALL(S)					
HC(0) ALWAYS HAS PRIORTY OVER HC(1)					
HURRY CALL No	CALL STAGE	CALL DELAY	CALL HOLD	CALL PREVENT	NOTES
CALL LIMIT	LOW				
	HIGH				

OPTION FILTER GREEN ARROW IN MANUAL			
TO APPEAR	NOT TO APPEAR	DEMAND DEPENDANT	INSERT TICK

PART TIME MODE	
SWITCH OFF IN STAGE(S)	

*

NORMAL ACTION IS A TOGGLE ACTION
(PRESS ONCE FOR ACTIVE, PRESS AGAIN TO MAKE INACTIVE

PUT TICK IF MOMENTARY ACTION IS REQUIRED
(ACTIVE ONLY WHILE BUTTON IS PRESSED)

IF SWITCH IS MOMENTARY OR NOT USED THEN THE ASSOCIATED LED IS AVAILABLE FOR OTHER PURPOSES

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EXTEND ALL RED FACILITY - (BY DETECTOR) CONSIDERED AS PART OF INTERGREENS

Stage to Stage transition for which ALL RED extension required					
UNIT No	From Stage	To Stage	Ext. Time	All Red Max	Associated Detector
1	1	3	2	6	AR1
2	1	4	2	6	AR1
3	1	5	2	6	AR1
4	2	3	2	6	AR1
5	2	4	2	6	AR1
6	2	5	2	6	AR1
7					

EXTEND ALL RED TIMINGS				
STREAMS	0	1	2	3
EXTENSION TIME (SECS)*				
ALL RED MAXIMUM TIME (SECS)*				

* Measured from point at which first phase in new stage would normally start its red-green transition period.

RED EXTENSION	ENTER CHOICE
MANUAL	
CLF	
UTC	
HC	
PRIORITY	
EMERGENCY	
VA	
FT	

For Choice, enter as follows:-
O = Operative
M = Extend to Maximum
N = No extensions
(Not approved for UK)

If table is left blank then all modes will be operative.

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MODE PRIORITY

ASSIGN NUMBER 1 = TOP PRIORITY
2 = SECONDY PRIORITY etc.
X = NOT USED

	PART TIME
	EMERGENCY VEHICLE
	HURRY CALL(S)
1	URBAN TRAFFIC CONTROL
2	SELECTED MANUAL CONTROL (1)
3	SELECTED FIXED TIME OR VA (2)
5	CABLELESS LINKING
4	MOVA
6	EITHER VEHICLE ACTUATED
	OR FIXED TIME

PERMANENTLY DISABLED MODES				
MODES TO BE DISABLED	STREAM			
	0	1	2	3
PART TIME (3)				
EMERGENCY VEHICLE				
HURRY CALL(S)				
SELECTED MANUAL CONTROL (3)				
URBAN TRAFFIC CONTROL				
SELECTED FIXED TIME OR VA				
CABLELESS LINKING				
BUS PRIORITY				
VEHICLE ACTUATED				
FIXED TIME				

✓ TICK IF REQUIRED TO BE DISABLED.

MANUAL FACILITY ENABLED	
ALWAYS	✓
WHEN HANDSET PLUGGED IN	
WHEN "MND" COMMAND ENTERED	

FACILITY SWITCH POSITION	NOT OPERATIVE INSERT (X)
VA	
FIXED TIME	
CLF	

(3) THESE MODES MAY ONLY BE DELETED ON ALL MODES SIMULTANEOUSLY

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FIXED CYCLE TIME MODE STAGE SEQUENCE								
Phase	A	B	C	D	E	F	G	H
Stage	1st	2nd	3rd	4th	5th	6th	7th	8th
Time								

OR

ALTERNATIVELY FIXED CYCLE RUNNING TO CURRENT TIMETABLED MAXIMA																
Phase	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Demand Dependent					✓	✓										
Phase	Q	R	S	T	U	V	Q	W	X	Y	Z					
Demand Dependent																

MANUAL SELECTION		
Button No.	Stage(s) called	Street Names - If Required
ALL RED	All Red Stage(s)	
1	1	York Road
2	2	York Road E/B w. RT
3	3	Peds
4	4	Scholes Lane

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URBAN TRAFFIC CONTROL

USE OF BITS | OTU MANUFACTURER & TYPE:-

Bit No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Control	SO	F1	F2	F3	F4	F5	D2	D3	D4	D5					GO		
Reply	MC	G1	G2	G3	G4	G5	DR2	DR3	DR4	DR5			LF1	LF2	MO	DF	
Secret Sign																	Related Phase/Stage
Secret Sign with security																	Related Phase/Stage
Bit No	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	
Control	DS1	DS2	DS3	DS4	SF1	SF2	SF3	SF4									
Reply									OC1	OC2	OC3	OC4	FL1	FL2	FL3	FL4	
Secret Sign																	Related Phase/Stage
Secret Sign with security																	Related Phase/Stage

Notes:- Control: SF Special facility used for Stage Holds, DS used for Dynamic Dataset changes
Replies: SC is Scoot bit (VS), FL is Vehicle Count bit (VC), OC is Vehicle Occupancy Bit (VO), Q is Queue loop, MO to reply when Mova is Offline.

0141 CONTROL AND REPLY BIT NOTATIONS

CONTROL BITS

Force (F1,F2,etc)
Stage Demand (D1,D2,etc)
Common Demand (DX)
SW FAC (SF1,SF2)
CLF Group Sync (SG)
RTC Sync (TS or TS1)
Fall Back Mode Select (FM)
Lamps On (LO)
Local Link Inhibit (LL)
Transmission Confirm (TC)
Solar Switch Over-ride (SO)
Pedestrian Demand (PX)
Hold Vehicle Stage (PV)
Solar Switch Over-ride (SO)

REPLY BITS

Stage Confirm (G1,G2,etc)
Stage Dem Confirm (DR1,DR2,etc)
SW FAC Confirm (SC1,SC2,etc)
CLF Group Sync Confirm (CS)
CLF Group 0 Confirm (GR1)
RTC Sync Confirm (CC or CC1)
Fall Back Confirm (FC)
Lamps Off Confirm (LE)
Hurry Call Confirm (HC)
Manual Control Mode (MC)
Controller Fault (CF)
Detector Fault (DF)
Remote Reconnect (RR)
One or more Signal Lamps Failed (LF)
Pedestrian Inhibit by Lamp Monitor (PI)
Pedestrian Reply (GX)

UTC CONTROL REQUIRED WHEN:-

✓ Force Bits Present
Force Bits and TC/TO Present
TC/TO Present

B.T. Termination requirements are:-

B.T. Line Number:-

OPTION 1 (MCE 0105/0106)
OPTION 2 (TCD316) (delete one)

INSERT PHASES OR STAGES TO BE DEMANDED, IF DEMANDS TO BE UNLATCHED ADD #

D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11
	D	E	C	G						
		F		H						

INSERT PHASES TO BE EXTENDED

D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11
	D		C	G						
				H						

INSERT PHASES OR STAGES FOR REPLY

DR1	DR2	DR3	DR4	DR5	DR6	DR7	DR8	DR9	DR10	DR11
	2	3	4	5						

DEMAND DEPENDENT FORCES INSERT DEMANDS TO BE CONSIDERED

F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
	D	E	C	G						
		F		H						

NOTES:-

1. Hash (#) indicates demand dependent force

2. Asterisk (*) denotes Control or Reply associated with separate Pelican

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UTC ONLY - MANUAL MODE, NO LAMP SUPPLY, LAMP FAILURES etc., CONFIRM					
TICK AS APPROPRIATE:-	G1/G2	MC	DF	LF1	LF2
MANUAL MODE SELECTED AND OPERATIVE	✓	✓			
NO LAMP POWER and/or SIGNALS OFF	✓				
DETECTOR FAULT			✓		
SELECT SWITCH NOT IN NORMAL POSITION		✓			
ANY LAMP FAILURE				✓	
2nd RED LAMP FAILURE ON ANY PHASE				✓	✓

Leeds UTC Lamp Monitoring Specification:

On any lamp failure the LF1 reply bit shall be returned.

On the 1st red lamp failure to any phase conflicting with a pedestrian phase, the LF1 reply bit shall be returned and the detector fault monitor lamp shall flash.

On any second red lamp failure to any phase LF2 shall be returned.

On any second red lamp failure to any phase conflicting with a pedestrian phase, the LF2 reply bit shall be sent and the controller shall switch off.

On any lamp failure there shall be no intergreen extensions.

Telent (Microsense Type) Timetable Data

Timetable Data			
No.	Day Type	Time	Event List No.
1	WKD	07:00:00	2
2	WND	07:00:00	1
3	WKD	09:30:00	1
4	FRI	15:00:00	3
5	MON-THU	15:45:00	3
6	WEK	19:00:00	4
7	WEK	07:00:00	9
8	WEK	22:00:00	10
9	WEK	00:00:00	11
10	WEK	00:00:01	12
11			
12			
13			
14			
15			
16			

Event List Data								
Event List No.	Event Action 1		Event Action 2		Event Action 3		Event Action 4	
	Type	Params	Type	Params	Type	Params	Type	Params
1	TTS	1						
2	TTS	2						
3	TTS	3						
4	TTS	4						
5	TTS	5						
6	TTS	6						
7	TTS	7						
8	TTS	8						
9	TEF	1=ON	Enable Bleepers					
10	TEF	1=OFF	Disable Bleepers					
11	TEF	2=ON	Enable Stage 1 Revert					
12	TEF	2=OFF	Disable Stage 1 Revert					
13								
14								
15								
16								

Day Types:-

- MON - Monday
- TUE - Tuesday
- WED - Wednesday
- THU - Thursday
- FRI - Friday
- SAT - Saturday
- SUN - Sunday
- WKD - Mon - Fri
- WND - Sat - Sun
- WEK - Every day
- XMO - All except Mon
- XTU - All except Tue
- XWE - All except Wed
- XTH - All except Thu
- XFR - All except Fri
- XSA - All except Sat
- XSU - All except Sun

Event Types:-

- TTS - timetable phase timing set
- TCF - timetable CLF
- TEF - timetable event flag
- TPT - timetable part-time
- TSF - timetable switched sign
- TDO - timetable detector override
- TDT - timetable detector timing set

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- MOVA LOGIC	
STREAM 1	
1 - 2	F2
1 - 3	F3
1 - 4	F4
1 - 5	F5
2 - 3	F3
2 - 4	F4
2 - 5	F5

3 - 4	F4
3 - 5	F5
3 - 1	F1
3 - 2	F2
4 - 5	F5
4 - 1	F1
4 - 2	F2
4 - 3	F3
5 - 1	F1
5 - 2	F2
5 - 3	F3
5 - 4	F4

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

$\overline{\text{EXT}}$ C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED "." - BOOLEAN "AND" "+" - BOOLEAN "OR"

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLELESS LINKING	
STREAM 1	
1 - 2	(($\overline{\text{F}}$1•F2)+ (F1•F2•GO•$\overline{\text{EXT}}$ $\overline{\text{B}}$)) • (DEM D + D2)
1 - 3	(($\overline{\text{F}}$1•F3) + (F1•F3•GO•$\overline{\text{EXT}}$ $\overline{\text{A}}$•$\overline{\text{EXT}}$ $\overline{\text{B}}$)) • (DEM E + DEM F + D3)
1 - 4	(($\overline{\text{F}}$1•F4) + (F1•F4•GO•$\overline{\text{EXT}}$ $\overline{\text{A}}$•$\overline{\text{EXT}}$ $\overline{\text{B}}$)) • (DEM C + D4)
1 - 5	(($\overline{\text{F}}$1•F5) + (F1•F5•GO•$\overline{\text{EXT}}$ $\overline{\text{A}}$•$\overline{\text{EXT}}$ $\overline{\text{B}}$)) • (DEM H + D5)
2 - 3	(($\overline{\text{F}}$2•F3) + (F2•F3•GO•$\overline{\text{EXT}}$ $\overline{\text{A}}$•$\overline{\text{EXT}}$ $\overline{\text{D}}$)) • (DEM E + DEM F + D3)
2 - 4	(($\overline{\text{F}}$2•F4) + (F2•F4•GO•$\overline{\text{EXT}}$ $\overline{\text{A}}$•$\overline{\text{EXT}}$ $\overline{\text{D}}$)) • (DEM C + D4)
2 - 5	(($\overline{\text{F}}$2•F5) + (F2•F5•GO•$\overline{\text{EXT}}$ $\overline{\text{A}}$•$\overline{\text{EXT}}$ $\overline{\text{D}}$)) • (DEM H + D5)
2 - 1	VIA 5
3 - 4	($\overline{\text{F}}$3•F4) • (DEM C + D4)
3 - 5	($\overline{\text{F}}$3•F5) • (DEM H + D5)
3 - 1	$\overline{\text{F}}$3•F1
3 - 2	($\overline{\text{F}}$3•F2) • (DEM D + D2)
4 - 5	(($\overline{\text{F}}$4•F5) + (F4•F5•GO•$\overline{\text{EXT}}$ $\overline{\text{C}}$)) • (DEM H + D5)
4 - 1	$\overline{\text{F}}$4•F1 + F4•F1•GO•$\overline{\text{EXT}}$ $\overline{\text{C}}$
4 - 2	(($\overline{\text{F}}$4•F2) + (F4•F2•GO•$\overline{\text{EXT}}$ $\overline{\text{C}}$)) • (DEM D + D2)
4 - 3	(($\overline{\text{F}}$4•F3) + (F4•F3•GO•$\overline{\text{EXT}}$ $\overline{\text{C}}$)) • (DEM E + DEM F + D3)
5 - 1	$\overline{\text{F}}$5•F1 + F5•F1•GO•MIN H
5 - 2	(($\overline{\text{F}}$5•F1) + (F5•F1•GO•MIN H)) • (DEM D + D2)
5 - 3	(($\overline{\text{F}}$5•F3) + (F5•F3•GO•MIN H)) • (DEM E + DEM F + D3)
5 - 4	(($\overline{\text{F}}$5•F3) + (F5•F3•GO•MIN H)) • (DEM C + D4)

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

$\overline{\text{EXT}}$ C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED “.” - BOOLEAN “AND” “+” - BOOLEAN “OR”

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- VEHICLE ACTUATED / FIXED TIME	
STREAM 1	
1 - 2	(MAX B + $\overline{\text{EXT}}$ $\overline{\text{B}}$) • DEM D
1 - 3	(MAX A + $\overline{\text{EXT}}$ $\overline{\text{A}}$) • (MAX B + $\overline{\text{EXT}}$ $\overline{\text{B}}$) • (DEM E + DEM F)
1 - 4	(MAX A + $\overline{\text{EXT}}$ $\overline{\text{A}}$) • (MAX B + $\overline{\text{EXT}}$ $\overline{\text{B}}$) • DEM C
1 - 5	(MAX A + $\overline{\text{EXT}}$ $\overline{\text{A}}$) • (MAX B + $\overline{\text{EXT}}$ $\overline{\text{B}}$) • DEM H
2 - 3	(MAX A + $\overline{\text{EXT}}$ $\overline{\text{A}}$) • (MAX D + $\overline{\text{EXT}}$ $\overline{\text{D}}$) • (DEM E + DEM F)
2 - 4	(MAX A + $\overline{\text{EXT}}$ $\overline{\text{A}}$) • (MAX D + $\overline{\text{EXT}}$ $\overline{\text{D}}$) • DEM C
2 - 5	(MAX A + $\overline{\text{EXT}}$ $\overline{\text{A}}$) • (MAX D + $\overline{\text{EXT}}$ $\overline{\text{D}}$) • DEM H
2 - 1	VIA 5
3 - 4	MAX E • MAX F • DEM C
3 - 5	MAX E • MAX F • DEM H
3 - 1	MAX E • MAX F • (DEM A + DEM B)
3 - 2	MAX E • MAX F • DEM D
4 - 5	(MAX C + $\overline{\text{EXT}}$ $\overline{\text{C}}$) • DEM H
4 - 1	(MAX C + $\overline{\text{EXT}}$ $\overline{\text{C}}$) • (DEM A + DEM B)
4 - 2	(MAX C + $\overline{\text{EXT}}$ $\overline{\text{C}}$) • DEM D
4 - 3	(MAX C + $\overline{\text{EXT}}$ $\overline{\text{C}}$) • (DEM E + DEM F)
5 - 1	MIN H • (DEM A + DEM B)
5 - 2	MIN H • DEM D
5 - 3	MIN H • (DEM E + DEM F)
5 - 4	MIN H • DEM C

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

$\overline{\text{EXT}}$ C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED “.” - BOOLEAN “AND” “+” - BOOLEAN “OR”

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